



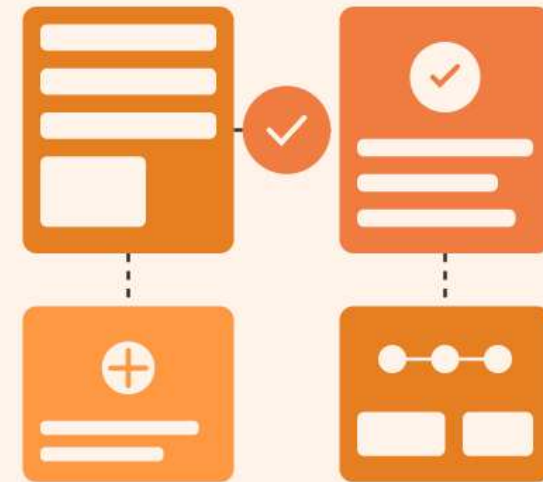
AI REAL ESTATE PLATFORM

Intelligent Solutions for Modern Real Estate

AI MODULAR SYSTEM

A modern, intelligent, modular AI backend for real estate websites

AI-Powered Real Estate Solutions





Core Modules

- **Recommender Module:** Suggests properties based on user input and property features
- **Analyzer Module:** Generates analytical insights about the real estate market
- **AI Agent (Chatbot) Module:** Conversational interface that collects preferences, triggers recommendations, and optionally books appointments

Design Principles

- ✓ Modular and independent services
- ✓ Tool-based orchestration (chatbot can call other modules)
- ✓ Professional, demo-ready, and easily integrated with .NET backend



COMPONENTS

- Content-Based Engine: Matches user preferences with property features
- Lightweight Hybrid Signals: Property popularity, recency, or featured status
- Ranking Engine: Combines scores and ranks properties

WHAT IT NEEDS

- Property dataset with features (price, location, bedrooms, bathrooms, area, type, amenities)
- Simulated property popularity/recency metrics
- User preference input from chatbot session

WHAT IT DOES

- Accepts user preferences as input
- Computes similarity between properties and preferences
- Combines content match and hybrid signals
- Returns top N recommended properties

TOOLS & TECHNOLOGIES

- Python, Pandas, NumPy
- Scikit-learn (vectorization & similarity calculation)
- FastAPI or Flask (for serving API)





COMPONENTS

- Data Aggregator: Groups and filters property data
- Trend Calculator: Computes average prices, median prices, monthly price changes, and listing counts
- Visualization Generator: Prepares charts and graphs for presentation

WHAT IT DOES

- Aggregates property data by region, type, or time
- Calculates trends and metrics (price changes, market growth)
- Returns results for frontend visualization

WHAT IT NEEDS

- Property dataset with price, date listed, location, and type

TOOLS & TECHNOLOGIES

- Python, Pandas, NumPy
- Matplotlib, Seaborn, Plotly
- FastAPI / Flask



COMPONENTS

- Intent Detection: Classifies user input (search, recommendation, booking, general query)
- Entity Extraction: Extracts details like location, budget, bedrooms, amenities
- Session Memory: Stores collected preferences during conversation
- Tool Caller: Calls recommender or analyzer modules based on detected intent
- Booking Handler: Collects appointment info and triggers backend booking API

WHAT IT NEEDS

- Access to recommender and analyzer modules
- Backend API endpoint for booking
- NLP tools for intent detection and entity extraction
- Session memory for multi-turn conversation

WHAT IT DOES

- Interprets user messages and detects intent
- Collects property preference information dynamically
- Calls hybrid recommender if recommendation intent is detected
- Provides analytics or other insights if requested
- Books appointments with property agents if requested

TOOLS & TECHNOLOGIES

- Python; spaCy or NLTK for NLP
- Optional LLM API (OpenAI GPT) for intent recognition
- FastAPI / Flask for chatbot API
- JSON / session storage for conversation context



RECOMMENDER

PURPOSE

Personalized property suggestions based on user preferences

KEY COMPONENTS

Content-Based Engine, Hybrid Signals, Ranking Engine

REQUIREMENTS

Property dataset, user preferences, popularity/recency metrics

TOOLS

Python, Pandas, Scikit-learn, FastAPI

ANALYZER

PURPOSE

Market insights and trend analytics for informed decisions

KEY COMPONENTS

Data Aggregator, Trend Calculator, Visualization Generator

REQUIREMENTS

Property dataset with price, date, location, and type data

TOOLS

Python, Pandas, NumPy, Matplotlib/Plotly, FastAPI

AI AGENT

PURPOSE

Conversational assistant and action trigger for user interactions

KEY COMPONENTS

Intent Detection, Entity Extraction, Session Memory, Tool Caller, Booking Handler

REQUIREMENTS

Access to Recommender/Analyzer, NLP tools, Booking API endpoint

TOOLS

Python, spaCy/NLTK, LLM API (optional), FastAPI, Session Storage

THANK YOU

Thank you for your interest in our AI Modular System for Real Estate. We're excited to help transform your platform with intelligent, scalable solutions.

Have questions about implementation, integration, or customization? Our team is ready to discuss how our modular AI architecture can meet your specific needs.

Ready to move forward?

Please reach out to us through your preferred communication channels to schedule a demo or technical consultation.

We look forward to collaborating with you to build the future of real estate technology.



LET'S CONNECT!

